



Local principal component transport for reduced-order reacting flow simulations

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ABSTRACT

In the development of reduced-order combustion models, data-driven principal component analysis (PCA) for dimensionality reduction has demonstrated its effective role in identifying and using low-dimensional manifolds to accelerate computing without losing fidelity. Recently, the PCA-based principal component transport (PC-transport) has been commonly adopted for reacting flow simulations, at substantially reduced computational cost associated with detailed chemical kinetics. However, the parametrization of the manifold is based on global variables (PC scores), which are common to the entire state-space, limiting the achievable degree of reduction offered by PC-transport. In this work, we present a new framework based on local PCA, wherein the state-space is mapped using the local PC score variables (L-PC scores), whose transport equations are integrated over time. This *local PC-transport* model (LPC-transport) allows the definition of the solution variables to change over time and space, increasing the level of reduction and improving computational performance. The L-PC variables are obtained using an unsupervised clustering, based on Vector Quantization PCA (VQPCA), with an on-the-fly PCA-based classifier during the simulations. The model is validated on ignition of homogeneous ammonia/air, propane/air and n-heptane/air mixtures against detailed chemistry, showing that LPC-transport allows to significantly increase the size reduction using only a few clusters and achieve substantial computational savings with larger mechanisms (nearly 170x for n-heptane with 654 species). The model is also validated in a two-dimensional (2D) hot spot ignition simulation for a propane/air mixture, where a significant size reduction is achieved compared to the full dimensional simulation with detailed chemistry. The 2D case also demonstrates the changes in definition of the transported variables over time and space, such that the most appropriate variables are solved for at each point in the domain.

1. Introduction

Numerical simulations of reacting flows are constrained by mainly two factors: large dimensionality due to the number of chemical species, and stiffness associated to the wide spectrum of chemical timescales. Such burden can be alleviated by the use of tabulated manifolds, where the state-space is parameterized using a reduced number of variables, capable of describing the most important features of the reactive system. Accordingly, the complexity of multi-scale problems, such as those arising in chemically reacting systems, has led to the development of various methods that reduce the number of species and chemical reactions but retain a desired degree of accuracy. Those strategies can be classified into two groups, namely *dynamics-based reduction* and *dimensionality reduction*.

Dynamics-based reduction aims at reducing the dimension of the mathematical model by pruning out unimportant species and reactions

while retaining a mathematical model involving only ordinary differential equations (ODEs). This simplified model can still be stiff since, in general, it includes both slow and fast scales, albeit it is generally less stiff than the original model. The identification of slow and fast scales aims at producing the minimal network that allows an accurate prediction of a set of target species. By construction, it is expected that only the target species will be accurately reproduced, while large errors can possibly affect the non-targeted species [1]. Various techniques have been proposed to simplify chemical mechanisms and reduce temporal stiffness. These can be classified as intuitive (such as steady-state approximation (SSA)), automated (computational singular perturbation (CSP) [2]), data-based (CSP with Jacobian pre-computed by data [3]), or data-based nonlinear using autoencoders [4].

Dimensionality reduction techniques, on the other hand, can be classified as intuitive (such as path flux analysis [5]), automated (directed

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relation graph (DRG) [6] and DRG with error propagation [7]), data-based linear (principal component analysis (PCA) [8]), or data-based nonlinear (autoencoders [4]). Among data-driven approaches, empirical manifolds based on PCA have received a wide interest due to their ability to identify low-dimensional subspaces based on orthonormal basis vectors, and parameterize the reacting system using a reduced number of optimal variables [9–13]. If the original thermo-chemical state-space consists of Q variables, PCA will provide a truncated set of q ($q < Q$) principal components (PCs) - which are linear combinations of the original variables — containing most of the information present in the original data. Therefore, a PCA-based reduced-order model (ROM) offers the ability to reduce the dimensionality of the system, accelerating the computation accordingly. In the *PC-transport* approach [10], the principal components scores (i.e. the projection of the thermo-chemical state-space on the PCA basis) are directly transported in the flow solver instead of the original species. However, special attention should be given to the PCs source terms accuracy: a linear projection of the source terms can strongly limit the achievable size reduction, thus requiring nonlinear regression techniques to further reduce the number of components to transport [13,14]. In all the aforementioned successful demonstrations of PCA-based ROMs of chemically reactive flows, the PC scores are defined using a global PCA basis matrix that is unique for the entire manifold. This global definition of the PC scores limits the achievable degree of reduction, especially in highly nonlinear data sets where a large number of scores is needed for an acceptable accuracy.

The present work seeks to enhance the PC-transport model using a local approach, where the PCA low-dimensional manifold is divided into clusters using an unsupervised algorithm based on Vector Quantization PCA (VQPCA), a subsequent PCA analysis is performed separately in each cluster providing an optimal local basis in each of them, and thereafter the local scores are transported instead of the global scores, allowing for a higher degree of reduction and computing performance. The local aspect of the approach resides in the fact that the chemical manifold in sub-divided into smaller clusters in the phase-space, allowing for fewer scores needed in each one of them as the clusters are less nonlinear than the global manifold. This new approach, called *local PC-transport*, is first demonstrated in zero-dimensional (0D) homogeneous reactors for different fuels with increasing level of chemical complexity, and subsequently in a two-dimensional (2D) hot spot ignition simulation with turbulent velocity fluctuations.

2. Methodology

2.1. PC-transport

PCA, as a data-driven statistical technique, allows for an alternative linear representation of the state-space by identifying a reduced number of optimal variables containing most of the data variability [9]. Starting from a data set $\mathbf{X}$ of dimension $(n \times Q)$, with n samples of Q independent variables, PCA provides an approximation based on q ($q < Q$) linear correlations:

$$\mathbf{X} \approx \mathbf{X}_q = \mathbf{Z}_q \mathbf{A}_q^T \quad (1)$$

where $\mathbf{X}_q$ ($n \times Q$) is the approximation of $\mathbf{X}$ based on the first q eigenvectors, $\mathbf{Z}_q$ ($n \times q$) are the principal components scores and $\mathbf{A}_q^T$ ($q \times Q$) is the matrix of the first q eigenvectors, i.e. the truncated basis matrix (the superscript T denotes the transposed). In this work, $\mathbf{X}$ contains the species mass fractions ($\mathbf{X} = [\mathbf{Y}_1, \dots, \mathbf{Y}_{ns}]$) and it is assumed that $\mathbf{X}$ has been appropriately centered and scaled before PCA is carried out [15].

By projecting the original species transport equation onto $\mathbf{A}_q$, the PC-transport equation is obtained [10]:

$$\frac{\partial}{\partial t}(\rho \mathbf{z}) + \nabla \cdot (\rho \mathbf{u} \mathbf{z}) = \nabla \cdot \mathbf{J}_z + \mathbf{S}_z \quad (2)$$

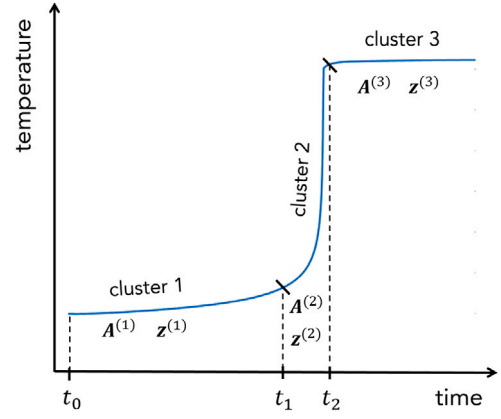


Fig. 1. Illustration of the local PC-transport concept on a 0D ignition curve.

where $\mathbf{z} = \mathbf{Z}_i^T$ represents an individual score (with $i = 1, \dots, q$), ρ is the density, $\mathbf{u}$ is the velocity, and $\mathbf{J}_z$ and $\mathbf{S}_z$ are the diffusive flux and chemical source terms of the principal components, respectively. $\mathbf{S}_z$ is defined as $\mathbf{S}_z = \mathbf{S}_y \mathbf{A}_q$ with $\mathbf{S}_y$ being the species chemical source terms, $\mathbf{J}_z = \rho \mathbf{D}_z \nabla \mathbf{z}$, with the score diffusion matrix $\mathbf{D}_z = \mathbf{A}_q^T \mathbf{D}_y \mathbf{A}_q$, and $\mathbf{D}_y$ is the diagonal matrix of the species diffusion coefficients. The calculation of $\mathbf{D}_z$ also requires some special treatment for the basis matrix $\mathbf{A}_q$ as discussed in [11], where the authors proposed a rotation to $\mathbf{A}_q$ in order to ensure that $\mathbf{D}_z$ is diagonal (or at least diagonal dominant). However, the calculation of $\mathbf{J}_z$ can be simplified by projecting the species diffusion flux $\mathbf{J}_y = \rho \mathbf{D}_y \nabla \mathbf{y}$ directly onto $\mathbf{A}_q$:

$$\mathbf{J}_z = (\rho \mathbf{D}_y \nabla \mathbf{y}) \mathbf{D}^{-1} \mathbf{A}_q \quad (3)$$

where $\mathbf{D}$ is the vector of scaling coefficients used for the PCA analysis. Eq. (3) alleviates the need to rotate $\mathbf{A}_q$ and compute $\mathbf{D}_z$ explicitly, and also improves the performance of the algorithm as fewer operations are required. This approach is used in the present work.

2.2. Local PCA

In reacting flows, a large number of PCs may be required to properly describe the system due to the intrinsically multi-linear nature of PCA [16]. To overcome this limit, a local PCA approach was introduced by Parente et al. [9], where the data set is divided into k clusters using an unsupervised algorithm based on Vector Quantization (VQ). The VQPCA algorithm partitions the data into separate clusters by minimizing the reconstruction error in each of them. Thereafter, PCA is performed in each of the clusters separately to find an optimal locally-reduced representation. The main advantage of dividing the dataset into several clusters is the ability to predict the high degree of nonlinearities with fewer components. Details about the VQPCA algorithm can be found in [9].

2.3. Local PC-transport

Fig. 1 illustrates the idea behind local PC-transport on an ignition curve. In this example, the dataset has been divided into $k = 3$ clusters, each having its own basis matrix, $\mathbf{A}_q^{(i)}$ ($i = 1, \dots, k$). Therefore, each cluster will also have its own definition of the scores, denoted $\mathbf{z}^{(i)}$. When switching from one cluster to the other, the definition of the transported variables will change. For instance, as illustrated in Fig. 1, at time $t = t_1$ the simulation will switch from cluster 1 to cluster 2, leading to a change in the definition of the resolved variables, from $\mathbf{z}^{(1)}$ to $\mathbf{z}^{(2)}$. In order to ensure a smooth transition, the idea proposed here is to impose the continuity of the physical state-space between two

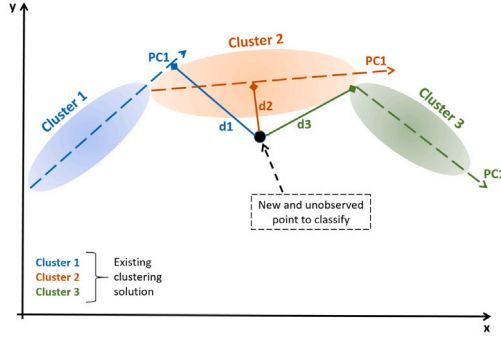


Fig. 2. Classification procedure via PCA reconstruction error [18].

consecutive clusters, i.e. $\mathbf{x}^{(1)} = \mathbf{x}^{(2)}$ at $t = t_1$. The latter can be rewritten using the scores definition:

$$\mathbf{z}^{(1)} \mathbf{A}_q^{(1)T} \mathbf{D}^{(1)} + \bar{\mathbf{X}}^{(1)} = \mathbf{z}^{(2)} \mathbf{A}_q^{(2)T} \mathbf{D}^{(2)} + \bar{\mathbf{X}}^{(2)} \quad (4)$$

where $\mathbf{D}^{(i)}$ and $\bar{\mathbf{X}}^{(i)}$ are the scaling and centering vectors in cluster i , respectively. By using this condition, the definition of the scores in the new cluster (i.e. the new variables to be integrated) can be retrieved:

$$\mathbf{z}^{(+)} = [(\mathbf{z}^{(*)} \mathbf{A}_q^{(*)T} \mathbf{D}^{(*)} + \bar{\mathbf{X}}^{(*)}) - \bar{\mathbf{X}}^{(+)}] \mathbf{D}^{(+)-1} \mathbf{A}_q^{(+)} \quad (5)$$

where $(*)$ is the current cluster index and $(+)$ the new cluster to transition to. The time-integration is then re-set at $t = t_1$ with $\mathbf{z}^{(2)}$ as its new initial state. The simulation carries on until it reaches $t = t_2$, where the scores for cluster 3 are computed by applying Eq. (5) between clusters 2 and 3. The cluster identification process requires an *on-the-fly* classification at each timestep, which is explained next.

2.4. Classification via PCA reconstruction error

During the simulation, in each cell and at each timestep, the local state-space vector $\mathbf{x}$ is projected onto each of the k local manifolds described by the pre-computed local basis matrices $\mathbf{A}_q^{(k)}$. Each cell is then classified to the cluster for which the PCA reconstruction error is minimized [17]. This procedure is shown graphically in Fig. 2. Given an existing clustering solution, the new state-space to be classified is projected onto each cluster (represented by their local PCA bases). The observation is then classified to the cluster minimizing the reconstruction error. If the cluster changes in a given cell, the new scores are computed using Eq. (5) and the time-integration proceeds in that cell with those new scores as the new initial state.

3. Test cases and numerical setup

The proposed local PC-transport model is applied to a 0D constant-pressure reactor model in which a system of nonlinear ODEs are integrated over time, as well as a 2D hot spot ignition case.

3.1. 0D ignition

For the 0D cases, three different reaction mechanisms were considered: a 38 species/260 reactions ammonia (NH_3) [19], a 114 species/1999 reactions propane (C_3H_8) [20] and a 654 species/5258 reactions n-heptane (nC_7H_{16}) [21] kinetic mechanisms. For all three cases, fuel/air mixtures at stoichiometric conditions ($\phi = 1.0$) and atmospheric pressure were simulated. The training datasets for the PCA analysis were generated using Cantera [22] with the same reactor configuration. A number of simulations were performed *off-line* by varying the initial temperature of the reactor from 1400 K to 1500 K in order to generate different ignition trajectories. The final datasets consisted of approximately 50000, 60000 and 64000 observations for

each of the state-space variables for the ammonia, propane and n-heptane cases, respectively. All of the data points were used collectively for the PCA analysis using Pareto scaling. All cases were run using a custom implementation of a 0D constant-pressure batch reactor in MATLAB employing Cantera and CVODE toolboxes. CVODE tolerances on the solution error were set to $\epsilon_{\text{rel}} = 10^{-4}$ and $\epsilon_{\text{abs}} = 10^{-10}$.

3.2. 2D ignition influenced by a hot spot

After validation of the reduced-order model for homogeneous auto-ignition of various fuels, ignition under the influence of a hot spot is considered in a two-dimensional (2D) setting under turbulence perturbation, such that spatial gradients and transport-chemistry interaction are present. A square computational domain is initialized with a uniform premixed mixture and turbulent velocity fluctuations. The hot spot is represented by a Gaussian-like spatial distribution of temperature, which has the form $T(x, y) = T_a + (T_m - T_a) \exp(-0.5r^2/(aL)^2)$, where T is the local temperature, T_m is the maximum value of temperature in the hot spot, T_a is the surrounding temperature, L is the length of the domain, $r = \sqrt{(x - L/2)^2 + (y - L/2)^2}$, and $a = 0.16$. A homogeneous, isotropic, turbulent velocity field was initialized by specifying the energy spectrum in the spectral space [23], having a root-mean-square velocity of 2.0 m/s and an integral length scale of 2.0 mm.

This case was simulated using the KARFS code [24] developed at KAUST, which solves the compressible Navier-Stokes, continuity, energy and species equations using a fourth-order, explicit Runge-Kutta scheme for time integration, while the spatial discretization is done using an eighth-order, non-dissipative, central difference operator. A stiff-ODE solver and a second-order operator-splitting algorithm are also available for stiff-chemistry problems. The unity-Lewis-number transport model was adopted, since the local PC-transport model is currently generated from data based on 0D configurations without inclusion of differential/preferential diffusion.

The 2D case is schematically illustrated in Fig. 3, showing the effect of the turbulent velocity fluctuations on the temperature field at $t = 4.8 \times 10^{-5}$ s. The computational domain is discretized with (500, 500) uniform grid points in the (x, y) directions, and the grid resolution is chosen as $\Delta x = \Delta y = 2 \times 10^{-5}$ m, such that the thermal thickness of a flame with the same initial T_a temperature is resolved with at least 20 grid points. The domain length in both directions is $L_x = L_y = 10$ mm and periodic boundary conditions are used. The surrounding temperature was $T_a = 1400$ K, while the peak hot spot temperature was set to $T_m = 1500$ K. The chosen fuel was propane, using the same mechanism and initial composition as for the 0D case described in Section 3.1.

The PCA training dataset used for the 2D case was the same as the propane 0D dataset described in Section 3.1. The local PCA analysis showed that using 12 PCs and 5 clusters should allow for a quasi-exact reconstruction of the state-space. Note that this reduction was adopted only for the chemistry sub-step of the simulation, in the context of an operator-splitting approach. In this study, the local PC-transport is demonstrated for 0D calculations and chemistry integration only. Therefore, for the transport step of the 2D simulation, the global scores were used and the prior analysis showed that at least 66 (global) PCs were required for an accurate representation of that manifold. In summary, 66 global scores were used for the transport part of the solver, while 12 local scores with 5 clusters were retained for the chemistry sub-step.

4. Results

4.1. 0D ignition cases

In this section, the performance of local PC-transport (hereafter labeled as 'L-PC') is compared with global PC-transport ('G-PC') and detailed chemistry, for various number of clusters. To this end, plots of

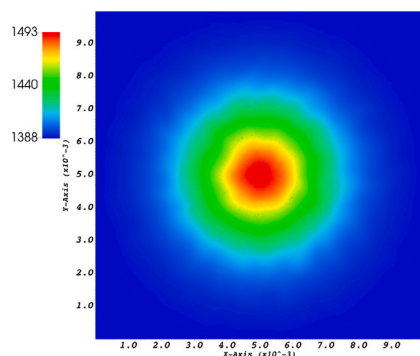


Fig. 3. 2D configuration and temperature distribution at $t = 4.8 \times 10^{-5}$ s.

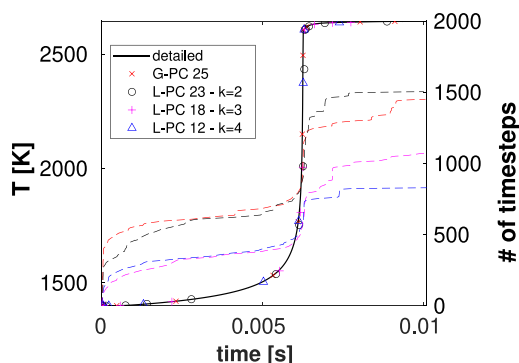


Fig. 4. Ammonia case — temperature solutions (left axis) for the detailed chemistry, G-PC and L-PC for various levels of clustering. The cumulative number of integration timesteps (right axis) is shown in dashed lines for the G-PC and L-PC using the same color as the corresponding symbols.

temperature, some selected species and cumulative number of timesteps are presented, while CPU time and ignition delay time (IDT) errors are also reported. For all figures in this subsection, plot markers are shown every 100 timesteps for better readability.

Figs. 4 and 5 show the time evolution of temperature and HO_2 mass fraction for the NH_3 case, comparing the detailed chemistry, the G-PC using 25 scores (representing a 35% reduction of the state-space), and the L-PC for 2 clusters with 23 scores (40% reduction), 3 clusters with 18 scores (53% reduction), and 4 clusters with 12 scores (69% reduction). Both the G-PC and the L-PC are able to accurately reproduce the detailed chemistry solution. The noticeable difference between them is the number of timesteps taken by the integrator, shown on the right y-axis in Fig. 4. The G-PC and L-PC with 2 clusters require approximately the same number of timesteps, which is significantly decreased when the number of clusters increases for L-PC. Indeed, increasing the number of clusters allows for an enhanced reconstruction of the chemical species in each of them, which in turn increases the accuracy of the species source terms (and, thereby, those of the scores), leading to a more stable time integration. This decrease in the number of timesteps naturally leads to a better overall performance and a decrease in CPU time, which is listed in Table 1. However, for this NH_3 case, L-PC remains slightly slower than detailed chemistry. A quantitative assessment is also presented in Table 1, listing the IDT error for the reduced models when compared to the detailed solution.

Figs. 6 and 7 show the time evolutions for temperature and CO mass fraction, respectively, for the C_3H_8 case. All the reduced models show excellent agreement with the detailed solution. It is also noticed that the G-PC model requires considerably more timesteps after the inflexion point of the ignition curve. This indicates that for this region of the solution, the reconstruction error on some species increases, which in

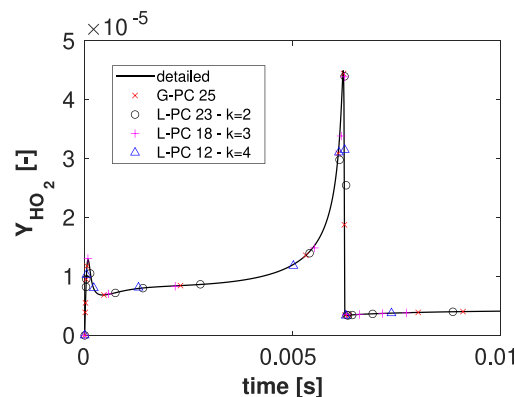


Fig. 5. Ammonia case - HO_2 mass fraction solutions for the detailed chemistry, G-PC and L-PC for various levels of clustering.

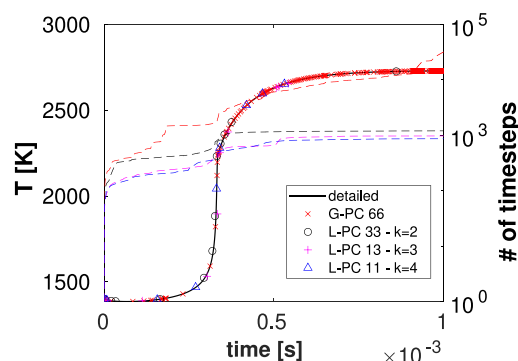


Fig. 6. Propane case — temperature solutions (left axis) for the detailed chemistry, G-PC and L-PC for various levels of clustering. The cumulative number of integration timesteps (right axis) is shown in dashed lines for the G-PC and L-PC using the same color as the corresponding symbols.

Table 1

Ammonia case — performance (CPU time) and accuracy (ignition delay time (IDT) error) comparison, for different number of clusters k . #PCs indicates the maximum achievable reduction.

model	#PCs (reduction %)	CPU time (s)	DT error (%)
G-PC	25 (35%)	1.5	0.15
L-PC - $k = 2$	23 (40%)	1.8	0.84
L-PC - $k = 3$	18 (53%)	1.1	0.76
L-PC - $k = 4$	12 (69%)	0.8	0.35
detailed	38 species	0.55	—

turn forces the integrator to take smaller timesteps in order to ensure a stable solution. On the other hand, a significant reduction in the number of timesteps is observed for all L-PC models (around 2 orders of magnitude), where the decomposition of the manifold into clusters leads to a higher species reconstruction accuracy. This also translates into improved performance and speed-up, decreasing the CPU time remarkably, as shown in Table 2. For this case, L-PC yields lower CPU times compared to the detailed chemistry case.

For the nC_7H_{16} case, the temperature and CH_2O solutions shown in Figs. 8 and 9, respectively, lead to similar observations as for the C_3H_8 case. Clustering and L-PC allow for a significant reduction in timesteps while ensuring accurate solutions. This is also confirmed quantitatively by the CPU time and IDT error presented in Table 3.

4.2. OD ignition — performance at low temperature

In order to further assess the performance of the L-PC model, an analysis is carried out at low temperature for the nC_7H_{16} case. The

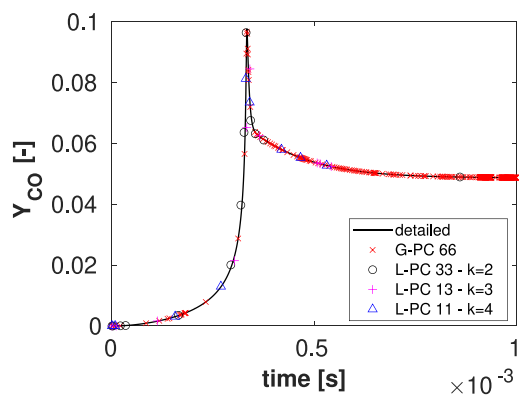


Fig. 7. Propane case - CO mass fraction solutions for the detailed chemistry, G-PC and L-PC for various levels of clustering.

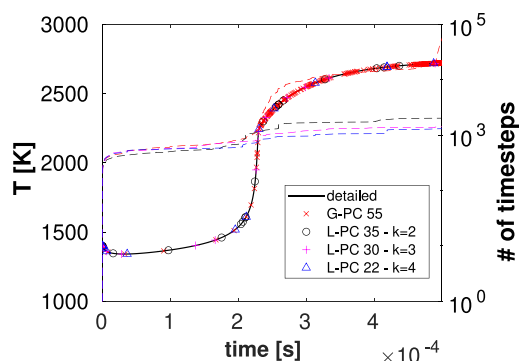


Fig. 8. n-Heptane case — temperature solutions (left axis) for the detailed chemistry, G-PC and L-PC for various levels of clustering. The cumulative number of integration timesteps (right axis) is shown in dashed lines for the G-PC and L-PC using the same color as the corresponding symbols.

Table 2

Propane case — performance (CPU time) and accuracy (ignition delay time (IDT) error) comparison, for different number of clusters k . #PCs indicates the maximum achievable reduction.

model	#PCs (reduction %)	CPU time (s)	IDT error (%)
G-PC	66 (40%)	990	0.05
L-PC - $k = 2$	33 (70%)	1.9	0.83
L-PC - $k = 3$	13 (89%)	1.23	1.16
L-PC - $k = 4$	11 (91%)	0.97	0.17
detailed	114 species	4.9	—

Table 3

n-Heptane case — performance (CPU time) and accuracy (ignition delay time (IDT) error) comparison, for different number of clusters k . #PCs indicates the maximum achievable reduction.

model	#PCs (reduction %)	CPU time (s)	IDT error (%)
G-PC	55 (92%)	14,537	0.16
L-PC - $k = 2$	35 (95%)	7.54	0.09
L-PC - $k = 3$	30 (96%)	6.31	0.71
L-PC - $k = 4$	22 (97%)	3.32	0.12
detailed	654 species	563	—

intent is twofold: as the initial temperature decreases, the ignition delay time increases which in turn allows for more error accumulation in the reduced model. This accumulation of error may lead to significant deviations. Moreover, at lower temperatures ($T < 1000$ K), the chemical kinetics become significantly more complex, especially due to low-temperature oxidation pathways. Therefore, in order to assess robustness over longer timescales and low-temperature chemistry, a

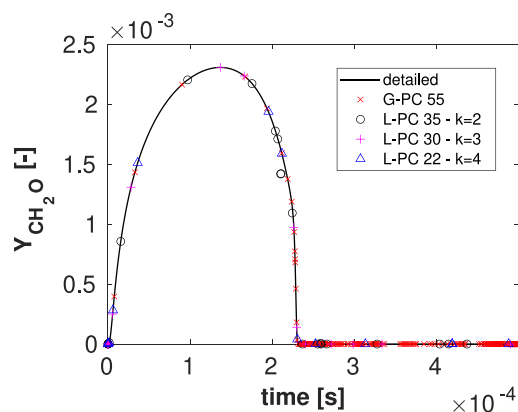


Fig. 9. n-Heptane case - CH₂O mass fraction solutions for the detailed chemistry, G-PC and L-PC for various levels of clustering.

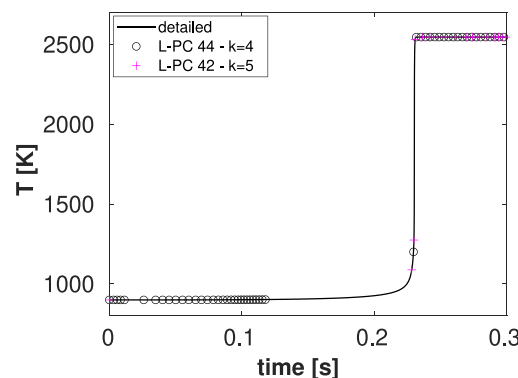


Fig. 10. n-Heptane case, $T_{init} = 900$ K - temperature solutions (left axis) for the detailed chemistry and L-PC for various levels of clustering.

Table 4

n-Heptane case, $T_{init} = 900$ K - performance (CPU time) and accuracy (ignition delay time (IDT) error) comparison, for different number of clusters k . #PCs indicates the maximum achievable reduction.

model	#PCs (reduction %)	CPU time (s)	IDT error (%)
L-PC - $k = 4$	44 (93%)	424	3.73
L-PC - $k = 5$	42 (94%)	266	0.92
detailed	654 species	427	—

new dataset was created for the nC₇H₁₆ case by varying the initial temperature from 800 K to 1000 K. This dataset consisted of approximately 175000 observations.

Fig. 10 shows the time evolution of temperature at an initial condition of 900 K, comparing the detailed chemistry solution with the L-PC model. It can be seen that for this low-temperature case, a larger number of clusters is required in order to achieve an accurate solution and performance gains. With 4 clusters, a reduction of the state-space size of 93% is achieved, but without notable CPU time reduction. On the other hand, using 5 clusters allows for a slightly higher reduction in size (94%), but with substantial CPU time decrease. The corresponding IDT error is reported in Table 4.

4.3. 2D ignition case

In this section, results obtained from the 2D computations using the detailed propane mechanism and the reduced L-PC model are compared for the ignition case under the influence of a Gaussian hot spot.

Figs. 11 and 12 show, respectively, the temperature and H₂O₂ mass fraction profiles corresponding to $t = 1.2 \times 10^{-4}$ s, extracted along

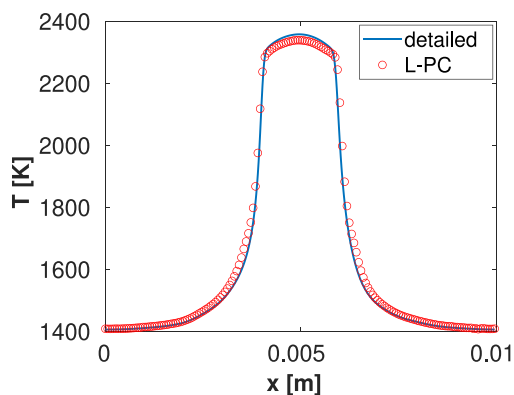


Fig. 11. Temperature profile extracted at $L_y/2$ for the detailed and L-PC solution at $t = 1.2 \times 10^{-4}$ s.

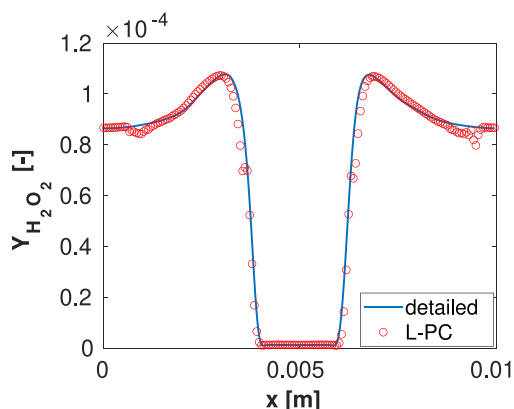


Fig. 12. H_2O_2 mass fraction profile extracted at $L_y/2$ for the detailed and L-PC solution at $t = 1.2 \times 10^{-4}$ s.

the x -direction at $L_y/2$, for the detailed and L-PC solutions. In the L-PC case, only 12 variables (scores) in 5 clusters are integrated for the chemistry part, instead of the 114 species for the detailed solution. It is observed in Fig. 11 that the L-PC model is able to accurately predict the temperature profile with a normalized root-mean-square error of $nrms = 7.12 \times 10^{-2}$, while some deviations from the detailed solution are seen on the profile of H_2O_2 in Fig. 12 ($nrms = 10.73 \times 10^{-2}$), especially near the boundaries of the domain. Although the error remains small, to some extent it can be attributed to the absence of transport information in the training dataset of the L-PC model. Another potential source of error may originate from the classifier. Indeed, if a cell is misclassified, it would lead to using incorrect PCA matrices for that state-space, which could also introduce errors.

Fig. 13 shows the 2D temperature field at $t = 1.2 \times 10^{-4}$ s, while Fig. 14 displays the corresponding distribution of clusters in the domain, i.e. it shows which part of the domain belongs to each of the 5 pre-computed clusters dividing the global chemical manifold. Those colored regions also highlight the zones where different sets of ODE equations are solved in the chemistry sub-step, with each set being the most appropriate for that particular state-space. Contrasting Fig. 14 with Fig. 12 reveals that the discrepancies observed in the H_2O_2 mass fraction profile occur mainly at the interfaces between clusters, especially between cluster #3 and #1, and between cluster #1 and #5. Therefore, this indicates that the transition process between clusters, as well as the classification technique, are consequential and could be further optimized. Finally, the L-PC simulation also showed improved computational performance, requiring 43% of the CPU time associated with the detailed simulation.

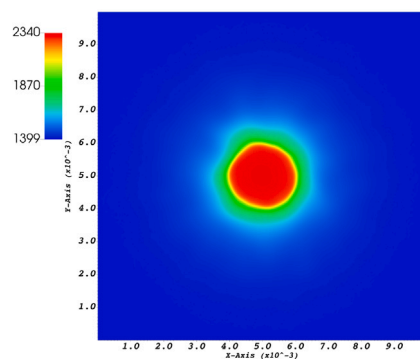


Fig. 13. Temperature field for the 2D case using L-PC at $t = 1.2 \times 10^{-4}$ s.

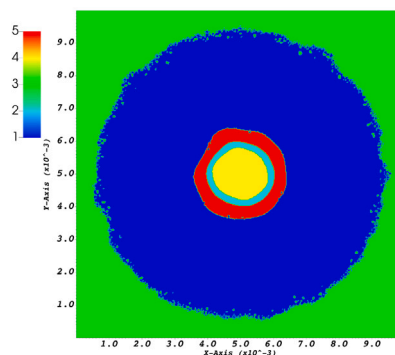


Fig. 14. Clusters index distribution for the 2D case using L-PC at $t = 1.2 \times 10^{-4}$ s.

5. Conclusion

The present work demonstrates the first application of the local principal component transport concept (Local PC-transport, L-PC), showing the potential of solving different sets of chemical ODEs at different time and space locations, and providing a method for a smooth transition between one cluster (one set of ODEs) to another one. The L-PC model is supported by Vector Quantization PCA (VQPCA) for the identification of the clusters, and by an *on-the-fly* PCA-based classification for the transition between clusters. The synchronization of the scores (the solution variables) between clusters is achieved by imposing the continuity of the physical state-space at the boundaries between clusters.

The L-PC model was successfully demonstrated in a 0D constant-pressure reactor setup for homogeneous NH_3/air , $\text{C}_3\text{H}_8/\text{air}$ and $\text{C}_7\text{H}_{16}/\text{air}$ mixtures. The achievable size reduction and computational speed-up were all increasing with the size of the reaction mechanism (nearly 170x acceleration for the n-heptane case), without compromising accuracy. The L-PC model also showed reasonable accuracy in a 2D setup (ignition under a hot spot with turbulence), although small discrepancies were found in some minor species. As initial implementation, the L-PC concept was limited to the chemistry sub-step of the operator-splitting method, while the transport part was solved using global PC (G-PC).

Further improvements of the model will focus on including transport and differential/preferential diffusion effects into the L-PC manifolds, in order to extend the model's ability to fully solve both transport and chemistry in a multi-dimensional simulation. The transition method between clusters will also be optimized to ensure a smoother transition and avoid spurious variations at cluster boundaries. Future work will also extend the proposed approach to more complex configurations and realistic cases.

Novelty and significance statement

Reduced-order models (ROMs) such as Principal Component Analysis based principal component transport (PC-transport) have been recently developed in the combustion community for dimensionality reduction. However, in those models the variables used to parameterize the manifold are defined on a global level, i.e. they are common to the entire state-space. This strongly limits their achievable degree of reduction. In this work, we present the first PC-transport model based on local variables, i.e. the integrated variables change over time and space, which not only increases the size reduction but also improves the overall performance. As such, this work is novel and of high impact to the community as high-fidelity simulations demand extensive computational resources. In this context, our proposed ROM model provides an efficient alternative to approximate high-fidelity models with lower computational requirements.

CRedit authorship contribution statement

Mohammad Rafi Malik: Designed research, Implemented the algorithms, Performed research, Analyzed data, Writing – original draft. **Francisco E. Hernández-Pérez:** Implemented the algorithms, Supervision, Writing – review & editing. **Hong G. Im:** Supervision, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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